

Principles of General Surgery and Critical Care

General Surgery — MBBS 4th and Final Year

Standard Core Curriculum for Medical Undergraduates • Comprehensive Clinical Reference

Curriculum Year	MBBS 4th and Final Year
Medical Subject	General Surgery
Target Audience	MBBS Undergraduates, Postgraduates, USMLE/PLAB Aspirants
Format	Fully Indexed Textbook with Chapter, Topic, and Page Referencing
Accreditation	Standard Medical Council Curriculum Guidelines

Chapter 1: Trauma, Shock and Resuscitation Protocols

Advanced Trauma Life Support (ATLS) Primary Survey and Thoracic Emergencies

The ATLS protocol establishes a prioritized ABCDE algorithm: Airway with cervical spine protection, Breathing with ventilation, Circulation with hemorrhage control, Disability (neurological evaluation GCS), and Exposure/Environmental control.

Immediate life-threatening thoracic trauma identified in primary survey: (1) Tension Pneumothorax presents with dyspnea, tracheal deviation away from affected side, absent breath sounds, hyperresonance, hypotension. Treatment is immediate needle decompression (14G cannula in 2nd ICS MCL or 5th ICS AAL) followed by tube thoracostomy (5th ICS MAL); (2) Cardiac Tamponade presents with Beck's Triad (hypotension, JVD, muffled heart sounds). FAST confirms diagnosis; treated by pericardiocentesis or pericardial window; (3) Massive Hemothorax (>1500 mL blood) requires chest tube and emergency thoracotomy if initial output >1500 mL or >200 mL/h for 2-4 hours.

Clinical Pearl: Never wait for an X-ray to treat a Tension Pneumothorax; it is a clinical diagnosis requiring immediate needle decompression followed by chest tube placement.

	Immediate Bedside Intervention	Definitive Management
hypotension	Needle thoracostomy (14G cannula, 2nd ICS MCL or 5th ICS AAL)	Tube thoracostomy
paradoxus	FAST ultrasound confirmation	Pericardiocentesis
	Large-bore chest tube (28-32 Fr)	Emergency thoracotomy

Chapter 1: Trauma, Shock and Resuscitation Protocols

Classification and Hemodynamic Management of Hemorrhagic and Septic Shock

Shock is systemic tissue hypoperfusion causing cellular hypoxia and lactic acidosis. ATLS classifies hemorrhage into four classes: Class I (<15% blood loss, <750 mL, normal vitals); Class II (15-30% loss, 750-1500 mL, tachycardia 100-120 bpm, normal SBP, narrow pulse pressure; crystalloid resuscitation); Class III (30-40% loss, 1500-2000 mL, tachycardia >120 bpm, hypotension, oliguria; requires blood + crystalloids); Class IV (>40% loss, >2000 mL, severe shock, negligible urine; requires Massive Transfusion Protocol 1:1:1 PRBC, FFP, Platelets).

Septic Shock is vasodilation and capillary leak from systemic infection. Surviving Sepsis Campaign bundle mandates within 1 hour: measure lactate, obtain blood cultures before antibiotics, administer broad-spectrum IV antibiotics, infuse 30 mL/kg balanced crystalloids for hypotension (MAP < 65 mmHg) or lactate >= 4 mmol/L, and apply vasopressors (Norepinephrine first-line) to maintain MAP >= 65 mmHg.

Clinical Pearl: Hypotension appears only in Class III shock (>30% blood loss); narrow pulse pressure and tachycardia are the earliest signs of Class II. Norepinephrine is first-line vasopressor for septic shock.

Heart Rate	Blood Pressure
< 100 bpm	Normal
100 - 120 bpm	Normal systolic, decreased pulse pressure
120 - 140 bpm	Decreased (Hypotension)
> 140 bpm	Profound hypotension, narrow PP

Chapter 2: Acute Abdominal Emergencies and Surgical Oncology

Acute Appendicitis: Pathogenesis, Alvarado Scoring and Surgical Management

Acute appendicitis is initiated by luminal obstruction of the vermiform appendix by a fecalith (adults) or lymphoid hyperplasia (children). Continued mucus secretion elevates intraluminal pressure, provoking ischemia, bacterial invasion, and gangrene.

Pain begins periumbilically (T10 dermatome visceral pain) then localizes to Right Iliac Fossa at McBurney's point (somatic peritoneal pain). Diagnostic signs: Rovsing sign (LLQ palpation causes RIF pain), Psoas sign (retrocecal appendix pain on hip hyperextension), Obturator sign (pelvic appendix pain on internal hip rotation).

Alvarado Score: Migration of pain (1), Anorexia (1), Nausea/vomiting (1), Tenderness in RIF (2), Rebound tenderness (1), Elevated temp >= 37.3°C (1), Leukocytosis >= 10,000 (2), Left shift (1). Score >= 7 strongly predicts appendicitis warranting appendectomy.

Clinical Pearl: Psoas sign indicates retrocecal appendicitis; Obturator sign indicates pelvic appendicitis. Alvarado score >= 7 warrants prompt appendectomy.

Alvarado Points	Physiological / Anatomical Mechanism
points total	Migration of pain to RIF (1), Anorexia (1), Nausea/Vomiting (1)
points total	Tenderness in RIF (2), Rebound tenderness (1), Elevated temperature >= 37.3°C (1)
points total	Leukocytosis > 10,000 (2), Neutrophilic band shift to left (1)
0 points	Score 1-4: Unlikely; Score 5-6: Observation/CT; Score 7-10: Appendectomy

Chapter 2: Acute Abdominal Emergencies and Surgical Oncology

Inguinal Hernias: Anatomy, Hesselbach Triangle and Repair Techniques

Inguinal hernias are classified into Indirect and Direct based on their relation to inferior epigastric vessels. Indirect Inguinal Hernia passes through the deep ring, lateral to inferior epigastric artery, within spermatic cord (patent processus vaginalis; most common in both sexes, can enter scrotum).

Direct Inguinal Hernia protrudes directly through Hesselbach's Triangle, medial to inferior epigastric vessels, due to transversalis fascia weakness (acquired in older men; rarely strangulates). Hesselbach's triangle is bounded by rectus abdominis (medial), inferior epigastric vessels (lateral), and inguinal ligament (inferior).

Deep ring occlusion test: examiner occludes the internal ring (1.25 cm above midinguinal point) while patient coughs. If hernia is controlled, it is indirect; if it emerges medial to the finger, it is direct. Standard repair is tension-free mesh hernioplasty (Lichtenstein technique).

Clinical Pearl: Indirect hernias are lateral to inferior epigastric vessels (patent processus vaginalis); Direct hernias are medial in Hesselbach's triangle (acquired fascial weakness).

Indirect Inguinal Hernia	Direct Inguinal Hernia
Lateral to vessels	Medial to vessels (within Hesselbach triangle)
Passes through deep inguinal ring into canal	Pushes directly through transversalis fascia
Frequently descends into scrotum	Rarely descends into scrotum
Higher risk due to narrow deep ring neck	Low risk due to broad, wide neck
Hernia controlled by occlusion of deep ring	Hernia still protrudes despite occlusion of deep ring